1. General Description

A: SPECIFICATION

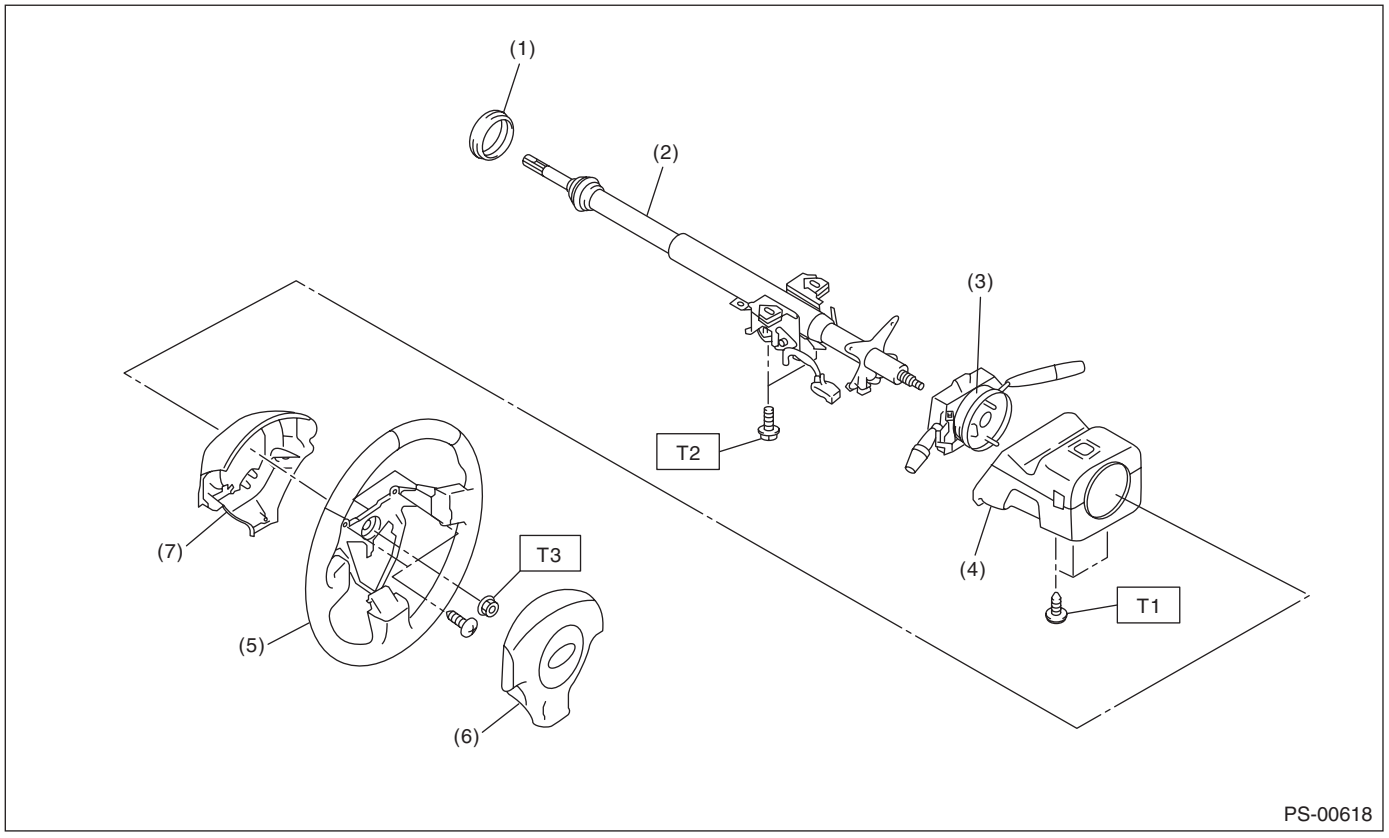
Model			Non-turbo	Turbo	
			2.5i, OUTBACK	WRX	STI
Whole system	Minimum turning radius m (ft)		5.4 (17.7)		5.7 (18.7)
	Steering angle (inside — outside)		34.5° — 30.3°		32.9° — 28.5°
	Steering wheel diameter mm (in)		375 (14.76)		375 (14.76)
	Steering rotations from lock to lock (while turning, from lock to lock. Gearbox)		3.0	2.7	2.6
Gearbox	Type		Rack and Pinion, Integral		
	Backlash		0 (Automatic adjusting)		
	Valve (Power steering system)		Rotary valve		
Pump (Power steering system)	Type		Vane pump		
	Oil tank		Installed on body		
	Specific output cm^3 (cu in)/rev.		7.2 (0.39)	7.2 (0.439)	
	Relief pressure kPa (kgf/cm^2, psi)		6,700 — 7,400 (68 — 75, 972 — 1,073)	7,350 — 8,036 (75 — 82, 1,067 — 1,165)	
	Hydraulic fluid control		Engine speed sensitive		
	Hydraulic fluid ℓ (US qt, Imp qt)		1,000 rpm: 7 (7.4, 6.2) 3,000 rpm: 5 (5.3, 4.4)	7.5 (7.9, 6.6)	
	RPM range rpm		700 — 9,000		
	Direction of rotation		Clockwise		
Hydraulic oil (Power steering system)	Capacity ℓ (US qt, Imp qt)	Oil tank	0.3 (0.3, 0.3)		
		Whole system	0.7 (0.7, 0.6)		

Steering wheel	Free play		mm (in)	17 (0.67)
Steering angle	Inner wheel	2.5i, OUTBACK, WRX		34.5°±1.5°
		STI		32.9°±1.5°
	Outer wheel	2.5i, OUTBACK, WRX		30.3°±1.5°
		STI		28.5°±1.5°
Steering shaft	Clearance between the steering wheel and column cover		mm (in)	4.0 (0.16)
Steering gearbox (Power steering system)	Sliding resistance		N (kgf, lbf)	400 (41, 90) or less
	Rack shaft play in the radial direction	Right-turn steering	mm (in)	0.19 (0.0075) or less
		Left-turn steering	mm (in)	Horizontal free play: 0.15 (0.0059) or less Vertical free play: 0.3 (0.012) or less
	Input shaft play	In radial direction	mm (in)	0.18 (0.0071) or less
		In axial direction	mm (in)	0.5 (0.020) or less
	Rotation resistance		N (kgf, lbf)	Maximum allowable value: 10.5 (1.1, 2.4) or less Difference between right and left sliding resistance: 20% or less
Oil pump (Power steering system)	Pulley shaft	Radial play	mm (in)	0.4 (0.016) or less
		Axial play	mm (in)	0.9 (0.035) or less
	Pulley	Ditch deflection	mm (in)	1.0 (0.039) or less
		Rotation resistance	N (kgf, lbf)	9.22 (0.94, 2.07) or less
	Regular pressure (Unloaded)		kPa (kgf/cm^2, psi)	981 (10, 142) or less
Steering wheel effort (Power steering system)	At standstill with engine idling on paved road		N (kgf, lbf)	31.4 (3.2, 7.1) or less
	At standstill with engine stalled on paved road		N (kgf, lbf)	294.2 (30, 66.2) or less

Recommended power steering fluid
SUBARU ATF or ATF DEXRON III

B: COMPONENT

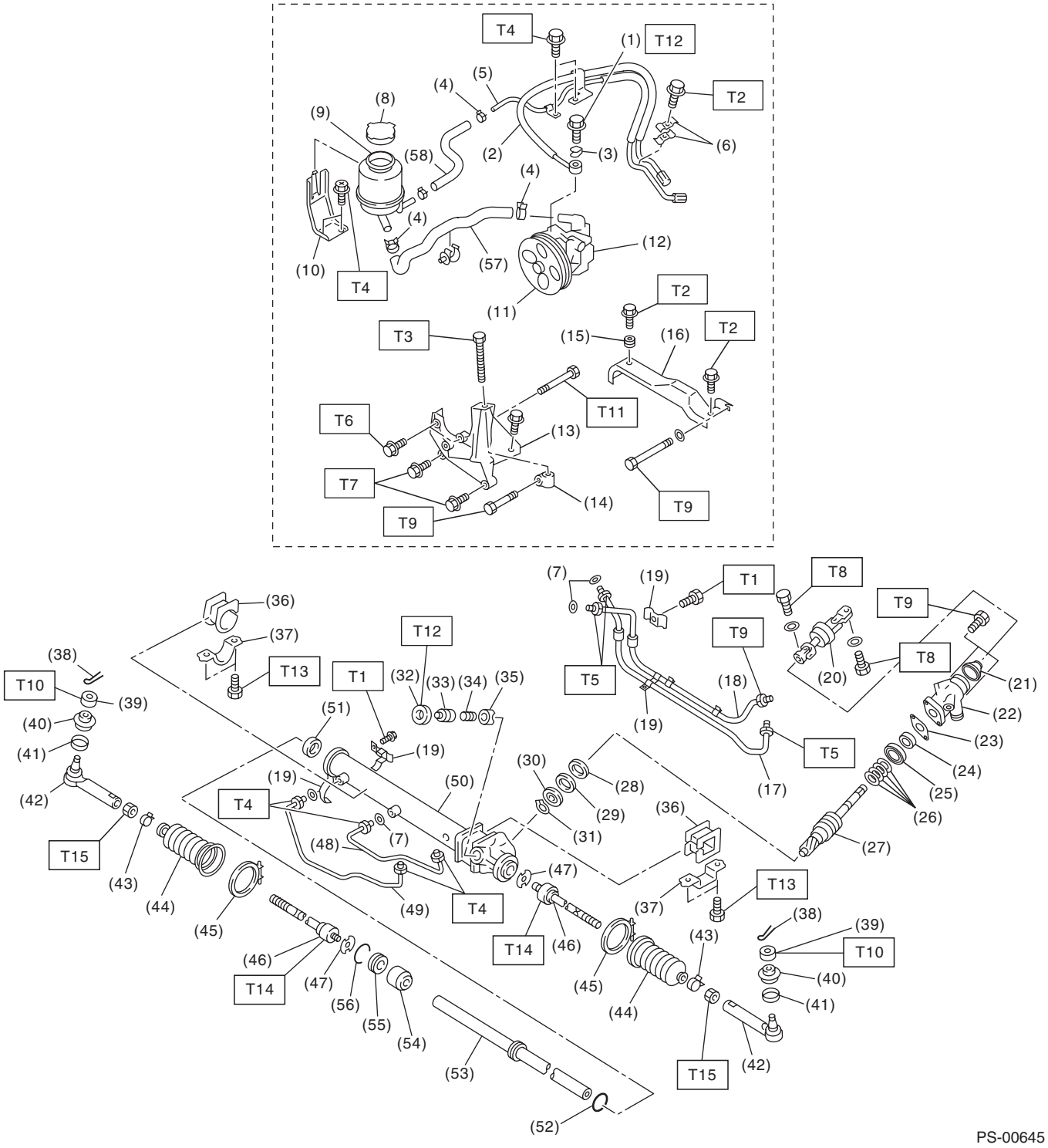
1. STEERING WHEEL AND COLUMN



(1)	Bushing	(5)	Steering wheel	***Tightening torque:N·m (kgf-m, ft-lb)***
(2)	Steering shaft	(6)	Airbag module	***T1: 1.2 (0.12, 0.9)***
(3)	Steering roll connector	(7)	Steering wheel lower cover	***T2: 25 (2.5, 18.1)***
(4)	Column cover			***T3: 45 (4.6, 33.2)***

2. POWER ASSISTED SYSTEM

Non-turbo model

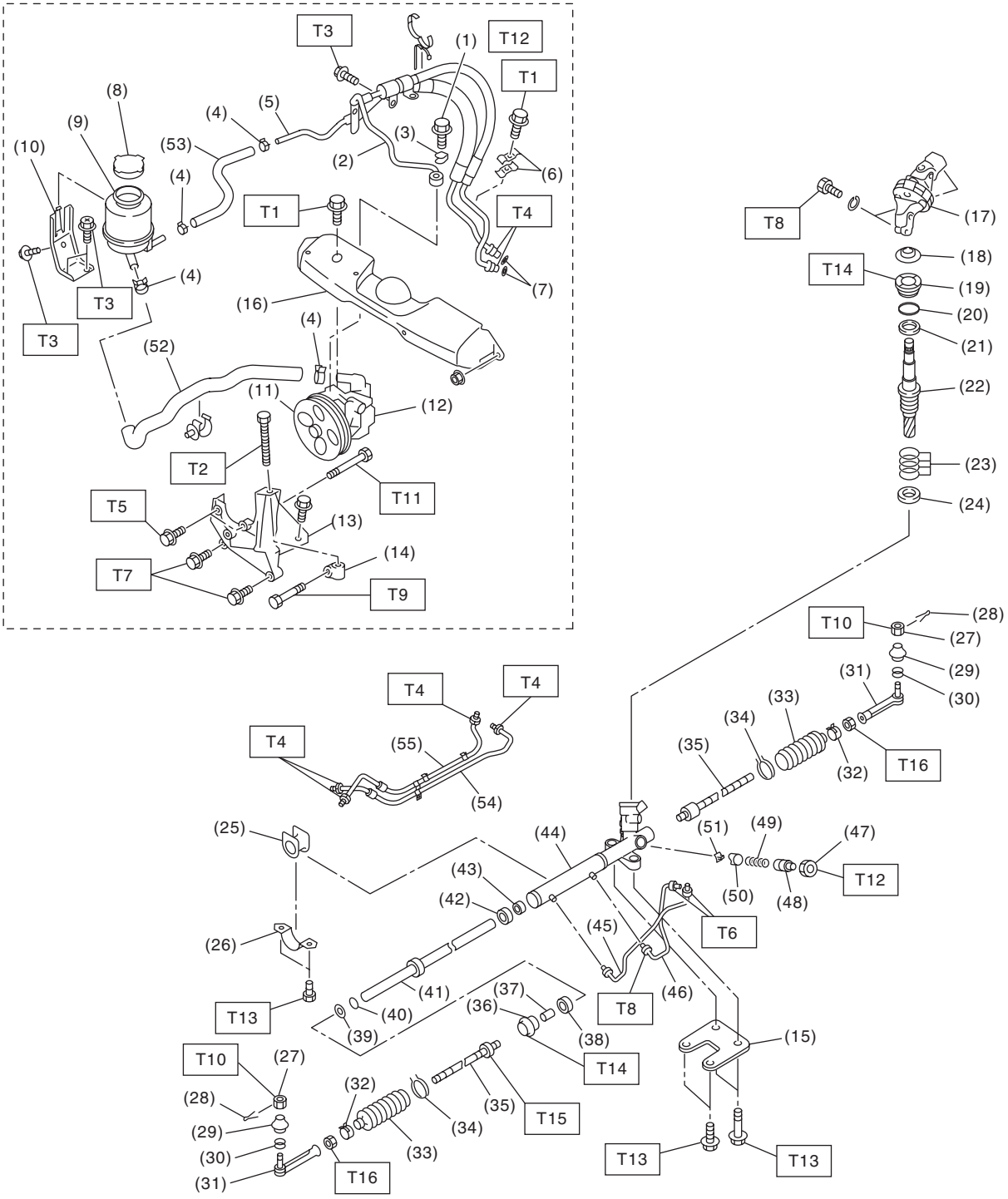


PS-00645

(1) Eyebolt
(2) Pipe C
(3) Gasket
(4) Clip
(5) Pipe D
(6) Clamp E
(7) O-ring
(8) Cap
(9) Reservoir tank
(10) Reservoir tank bracket
(11) Pulley
(12) Oil pump
(13) Bracket
(14) Belt tension nut
(15) Bushing
(16) Belt Cover
(17) Pipe E
(18) Pipe F
(19) Clamp plate
(20) Universal joint
(21) Dust seal
(22) Valve housing
(23) Gasket
(24) Oil seal
(25) Ball bearing
(26) Seal ring
(27) Pinion and valve ASSY
(28) Oil seal
(29) Back-up washer
(30) Ball bearing
(31) Snap ring
(32) Lock nut
(33) Adjusting screw
(34) Spring
(35) Sleeve
(36) Adapter
(37) Clamp
(38) Cotter pin
(39) Castle nut
(40) Dust cover
(41) Clip
(42) Tie-rod end
(43) Clip
(44) Boot
(45) Band
(46) Tie-rod
(47) Lock washer
(48) Pipe B
(49) Pipe A
(50) Steering body
(51) Oil seal
(52) Piston ring
(53) Rack
(54) Rack bushing
(55) Rack stopper
(56) Snap ring
(57) Suction hose
(58) Hose

Tightening torque:N·m (kgf-m, ft-lb)	
T1:	***6 (0.6, 4.3)***
T2:	***7.4 (0.75, 5.4)***
T3:	***8 (0.8, 5.8)***
T4:	***13 (1.3, 9.4)***
T5:	***15 (1.5, 10.8)***
T6:	***15.7 (1.6, 11.6)***
T7:	***22 (2.2, 15.9)***
T8:	***24 (2.4, 17.4)***
T9:	***25 (2.5, 18.1)***
T10:	***27 (2.75, 19.9)***
T11:	***37.3 (3.8, 27.5)***
T12:	***40 (4.1, 29.5)***
T13:	***60 (6.1, 44.3)***
T14:	***78 (8.0, 57.9)***
T15:	***83 (8.5, 61.5)***

Turbo model and STI model



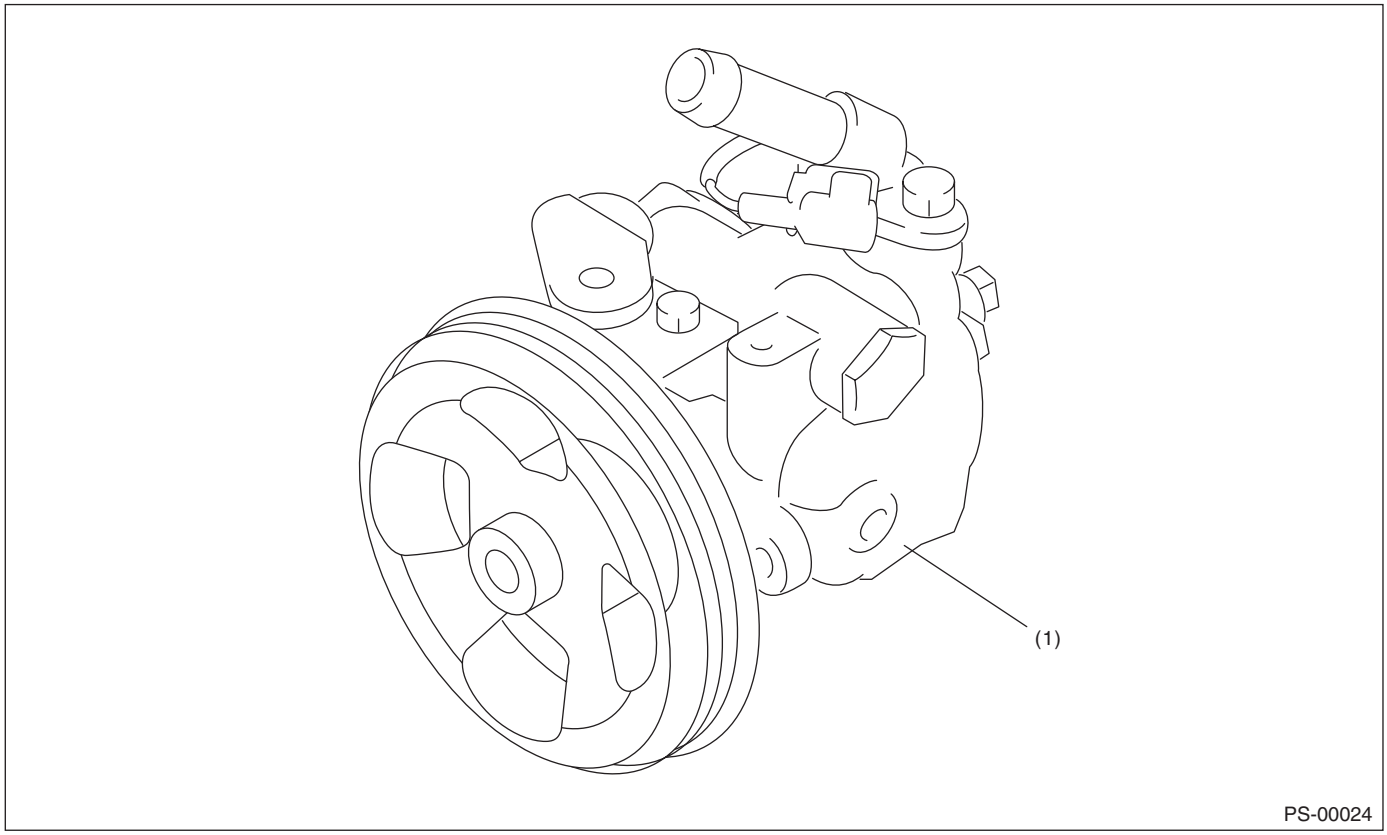
PS-00644

(1) Eyebolt
(2) Pipe C
(3) Gasket
(4) Clip
(5) Pipe D
(6) Clamp E
(7) O-ring
(8) Cap
(9) Reservoir tank
(10) Reservoir tank bracket
(11) Pulley
(12) Oil pump
(13) Bracket
(14) Belt tension nut
(15) Stiffener
(16) Belt Cover
(17) Universal joint
(18) Dust cover
(19) Plug
(20) O-ring
(21) Oil seal
(22) Pinion
(23) Seal ring
(24) Oil seal
(25) Adapter
(26) Clamp
(27) Castle nut
(28) Cotter pin
(29) Dust seal
(30) Clip
(31) Tie-rod end
(32) Clip
(33) Boot
(34) Wire
(35) Tie-rod
(36) Holder
(37) Bushing
(38) Oil seal
(39) Oil seal
(40) O-ring
(41) Rack
(42) Oil seal
(43) Back-up washer
(44) Steering body
(45) Pipe A
(46) Pipe B
(47) Lock nut
(48) Adjusting screw
(49) Spring
(50) Sleeve
(51) Sheet pad
(52) Suction hose
(53) Return hose
(54) Pipe G
(55) Pipe H

Tightening torque:N·m (kgf-m, ft-lb)	
T1:	***7.4 (0.75, 5.4)***
T2:	***8 (0.8, 5.8)***
T3:	***13 (1.3, 9.4)***
T4:	***15 (1.5, 10.8)***
T5:	***15.7 (1.6, 11.6)***
T6:	***20 (2.0, 14.5)***
T7:	***22 (2.2, 15.9)***
T8:	***24 (2.4, 17.4)***
T9:	***25 (2.5, 18.1)***
T10:	***27 (2.75, 19.9)***
T11:	***37.3 (3.8, 27.5)***
T12:	***40 (4.1, 29.5)***
T13:	***60 (6.1, 44.3)***
T14:	***64 (6.5, 47.0)***
T15:	***90 (9.0, 65.1)***
T16:	***85 (8.6, 62.2)***

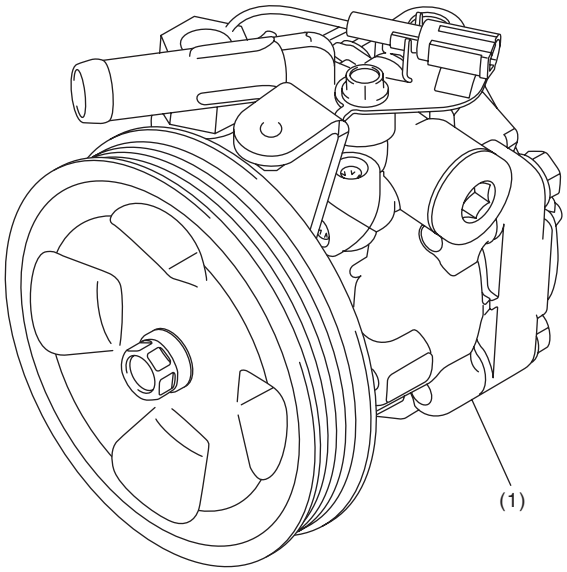
3. OIL PUMP

Non-turbo model



(1) Power steering oil pump ASSY

Turbo model and STI model



PS-00440

(1) Power steering oil pump ASSY

C: CAUTION

- Wear appropriate work clothing, including a cap, protective goggles and protective shoes when performing any work.
- Before removal, installation or disassembly, be sure to clarify the failure. Avoid unnecessary removal, installation, disassembly and replacement.
- Be careful not to burn your hands, because each part on the vehicle is hot after running.
- Use the genuine power steering fluid, grease etc. or the equivalent. Do not mix fluid, grease etc. of different grades or manufacturers.
- Be sure to tighten fasteners including bolts and nuts to the specified torque.
- Place shop jacks or rigid racks at the specified points.
- Before securing a part on a vise, place cushioning material such as wooden blocks, aluminum plate or cloth between the part and the vise.

D: PREPARATION TOOL

1. SPECIAL TOOL

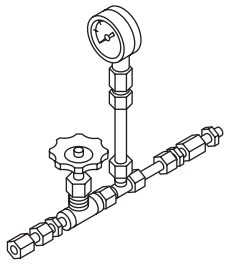
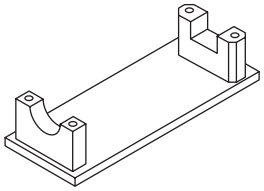
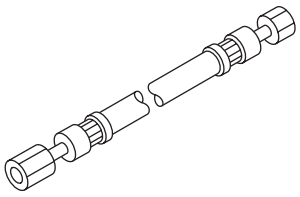
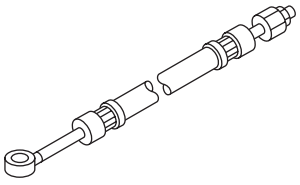
ILLUSTRATION	TOOL NUMBER	DESCRIPTION	REMARKS
 ST-925711000	925711000	PRESSURE GAUGE	Used for measuring oil pressure.
 ST-926200000	926200000	STAND	• Used when inspecting characteristic of gear-box assembly and disassembling it. • For non-turbo model.
 ST34099AC010	34099AC010	ADAPTER HOSE A	Used together with the PRESSURE GAUGE (925711000).
 ST34099AC020	34099AC020	ADAPTER HOSE B	Used together with the PRESSURE GAUGE (925711000).

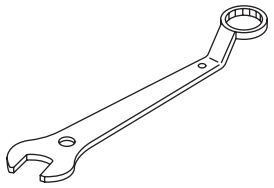
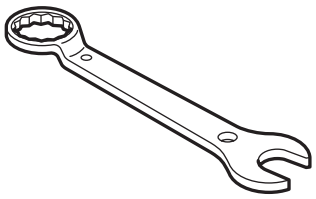
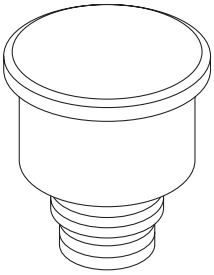
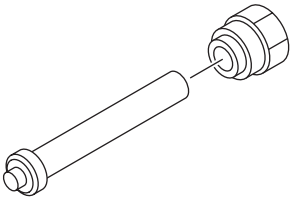
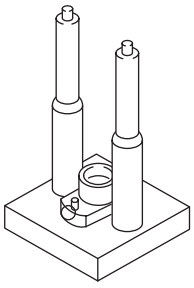
ILLUSTRATION	TOOL NUMBER	DESCRIPTION	REMARKS
 ST-926230000	926230000	SPANNER	For the lock nut when adjusting backlash of gearbox.
 ST34099PA100	34099PA100	SPANNER	Used when measuring the rotating resistance of gearbox assembly.
 ST34199AE040	34199AE040	OIL CHARGE GUIDE	Used for charging power steering fluid.
 ST-926420000	926420000	PLUG	When fluid leaks from pinion side of gearbox assembly, remove pipe B from valve housing, attach this tool and check oil leaking points.
 ST-926370000	926370000	INSTALLER A	• Used for installing the valve assembly into the valve housing assembly. • Used together with the STAND BASE (34099FA100).

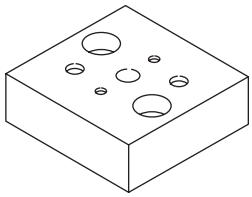
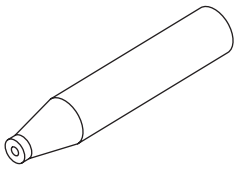
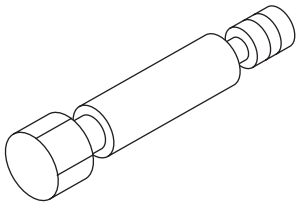
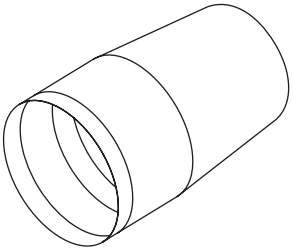
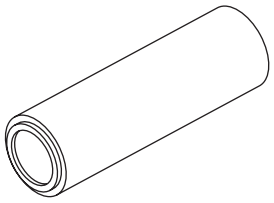
ILLUSTRATION	TOOL NUMBER	DESCRIPTION	REMARKS
 ST34099FA100	34099FA100	STAND BASE	Used for assembling power steering gearbox.
 ST-926390001	926390001	COVER & REMOVER ASSY	Used for assembling rack assembly.
 ST-926400000	926400000	GUIDE	• Used for installing rack bushing on the right side of the rack. • Used together with the GUIDE (927660000).
 ST-927660000	927660000	GUIDE	• Used for installing rack bushing on the right side of the rack. • Used together with the GUIDE (926400000).
 ST-927620000	927620000	INSTALLER B	• Used for installing oil seal of valve housing. • Used together with the INSTALLER A (926360000).

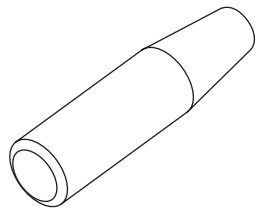
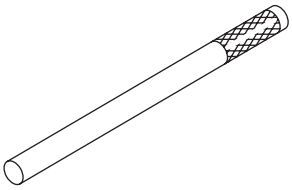
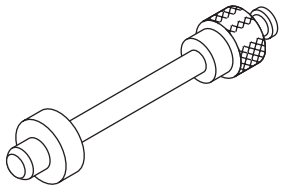
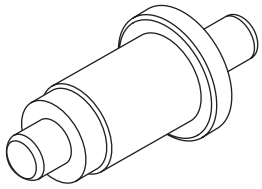
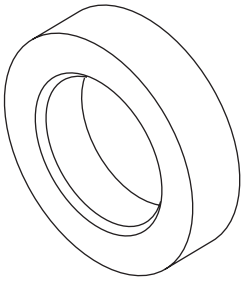
ILLUSTRATION	TOOL NUMBER	DESCRIPTION	REMARKS
 ST-926360000	926360000	INSTALLER A	• Used as a guide to install oil seal. • Used together with the INSTALLER B (927620000).
 ST34199AE050	34199AE050	OIL SEAL REMOVER	Used for removing the oil seal.
 ST34099FA110	34099FA110	INSTALLER	Used for installing the oil seal.
 ST34099FA120	34099FA120	INSTALLER & REMOVER SEAL	• Used for installing oil seal of valve housing. • Used together with the INSTALLER SEAL. (34099FA130) • Used for installing ball bearing of valve housing. • Used for removing oil seal and ball bearing from valve housing.
 ST34099FA130	34099FA130	INSTALLER SEAL	• Used for installing oil seal of valve housing. • Used together with the INSTALLER & REMOVER SEAL (34099FA120).

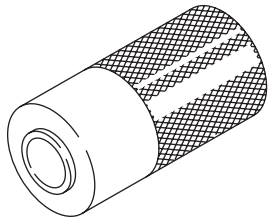
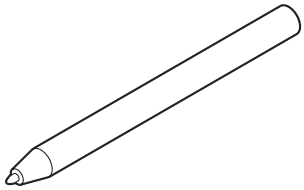
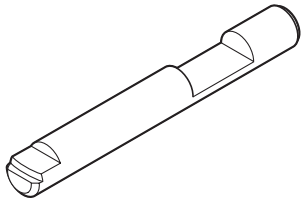
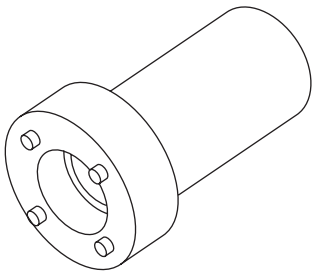
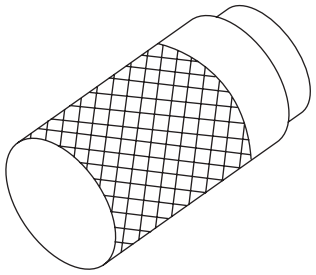
ILLUSTRATION	TOOL NUMBER	DESCRIPTION	REMARKS
 ST-927640000	927640000	INSTALLER B	Used for installing the ball bearing into housing.
 ST34099FA060	34099FA060	PUNCH HOLDER	• Used for crimping. • For turbo model and STI model.
 ST34099FA080	34099FA080	PUNCH	• Used for removing crimps. • For turbo model and STI model.
 ST34199AE090	34199AE090	PLUG WRENCH	• Used for removing the plug. • For turbo model and STI model.
 ST34199AE100	34199AE100	OIL SEAL PLUG REMOVER	• Used for removing the oil seal plug. • For turbo model and STI model.

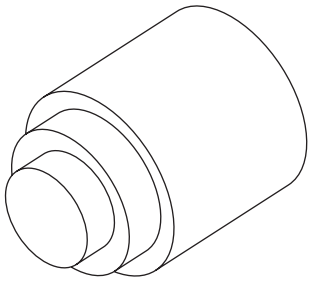
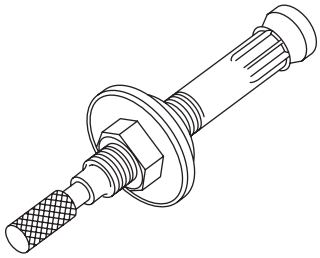
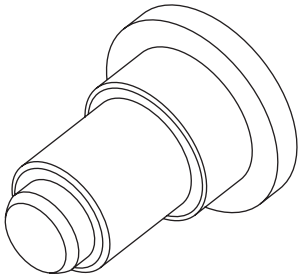
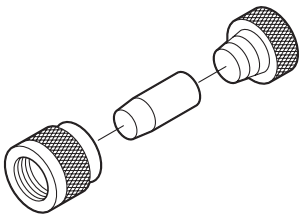
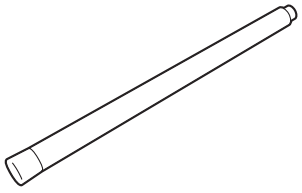
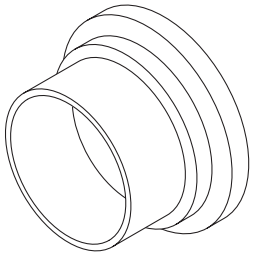
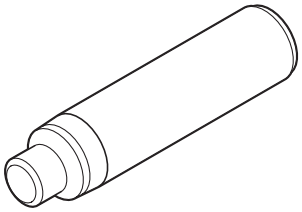
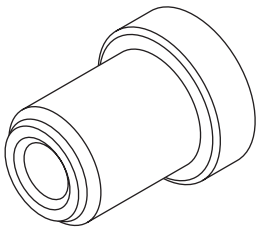
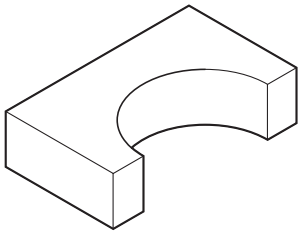
ILLUSTRATION	TOOL NUMBER	DESCRIPTION	REMARKS
 ST34199AE110	34199AE110	OIL SEAL PLUG INSTALLER	• Used for installing the oil seal plug. • For turbo model and STI model.
 ST34199AE120	34199AE120	GEARBOX OIL SEAL REMOVER	• Used for removing the gearbox oil seal. • For turbo model and STI model.
 ST34199AE130	34199AE130	GEARBOX OIL SEAL INSTALLER	• Used for installing the gearbox oil seal. • For turbo model and STI model.
 ST34199FE040	34199FE040	INSTALLER A, B, C	• Used for installing the oil seal to the rack assembly. • For turbo model and STI model.
 ST34199FE010	34199FE010	REMOVER	• Used for removing the back-up ring and oil seal. • For turbo model and STI model.

ILLUSTRATION	TOOL NUMBER	DESCRIPTION	REMARKS
 ST34199FE050	34199FE050	GUIDE	• Used for installing the rack and seal into housing assembly. • For turbo model and STI model.
 ST34199FE000	34199FE000	INSTALLER & REMOVER	• Used for removing and installing the rack oil seal (outer & inner). • For turbo model and STI model.
 ST34199FE060	34199FE060	INSTALLER	• Used for installing the rack oil seal (outer). • For turbo model and STI model.
 ST34199FE020	34199FE020	BASE	• Used for the support housing assembly. • For turbo model and STI model.

2. GENERAL TOOL

TOOL NAME	REMARKS
Spring scale	Used for measuring tightening torque.
Snap ring pliers	Used for removing and installing snap ring.
Dial gauge	Used for measuring the steering gearbox.
C clamp	Used for disassembling and assembling the steering gearbox of turbo model and STI model.